



CIS 422: Human Computer Interaction
Phase 2

FoodSaver

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Introduction

This report presents the pre-design analysis for the FoodSaver application, developed as part of the CIS422 Human-Computer Interaction course project. The following sections cover similar existing systems, relevant scientific

literature, the selected design framework, data collection method, user analysis, and task analysis.

Similar Systems

To inform the design of the FoodSaver application, four existing systems with related functionality were identified and analyzed. Each system was evaluated in terms of its purpose, usability characteristics, target users, and main tasks. This analysis supports a deeper understanding of the current landscape and highlights opportunities for FoodSaver to address unmet user needs.

System 1: NoWaste

NoWaste is a mobile application that aims to help users manage their food inventory at home. The application enables users to enter the food items along with their corresponding expiry dates and sends notifications to the users before the expiry dates. The application displays the food information in a graphical format.

Usability Characteristics	Simple and intuitive interface; clear navigation; visual organization of food items using color-coded expiration indicators; low cognitive load due to minimal data entry requirements.
Target Users	Household members including families, students, and individuals seeking to minimize food waste through organized inventory tracking.
Main Tasks	Add food items with expiration dates; monitor inventory status; receive expiration reminders; view food calendar.

Table 1: Usability characteristics, target users, and main tasks of NoWaste

System 2: FridgePal

FridgePal is a mobile application that allows a person to manage and track their refrigerator and pantry products. It assists in the classification of a food product, its expiry date, and managing the inventory. The application is intended to assist the user in knowing what is available in the food products at any given time.

Usability Characteristics	Category-based organization for easy navigation; clean and structured layout; supports quick item lookup; designed for everyday household use with minimal learning curve.
Target Users	Home cooks and individuals who prepare meals regularly and require an organized system for food inventory management.
Main Tasks	Add and remove food items; organize items by category; monitor expiration dates; manage household food inventory.

Table 2: Usability characteristics, target users, and main tasks of FridgePal

System 3: Yummly

Yummly is a recipe recommendation platform that suggests personalized meal ideas based on the user's available ingredients, dietary preferences, and cooking habits. It leverages machine learning to tailor recommendations and supports a rich media experience with step-by-step cooking instructions and integrated shopping lists.

Usability Characteristics	Personalized search and filtering; smooth and engaging interface; guided cooking mode; supports dietary filters (e.g., allergies, preferences); high visual appeal through professional food photography.
Target Users	Regular home cooks who seek inspiration and personalized recipe suggestions based on ingredients they already have.
Main Tasks	Search and browse recipes; filter by available ingredients or dietary restrictions; follow step-by-step cooking instructions; save favorite recipes.

Table 3: Usability characteristics, target users, and main tasks of Yummly

System 4: Too Good To Go

Too Good To Go is a sustainability-focused mobile application that connects users with local restaurants and food retailers offering surplus food at reduced prices. The application aims to reduce commercial food waste by facilitating the redistribution of unsold food items to consumers before they are discarded.

Usability Characteristics	Straightforward browsing and reservation interface; location-based discovery; real-time availability updates; simple checkout process designed for ease of use.
Target Users	Environmentally conscious consumers who want to reduce food waste and access affordable surplus food from nearby establishments.
Main Tasks	Browse available surplus food offers; reserve food packages; complete in-app purchase; pick up reserved items from the establishment.

Table 4: Usability characteristics, target users, and main tasks of Too Good To Go

Summary

The four systems mentioned above show a clear interest in food management and sustainable mobile applications. Although NoWaste and FridgePal are mainly centered around inventory management, and Yummly is more focused on recipe suggestions, none of the above-mentioned systems provide a comprehensive solution that integrates food inventory management, expiration dates, recipe suggestions based on ingredients, and social interaction in one application. FoodSaver is meant to close this gap by integrating all these aspects in one accessible interface. FoodSaver also introduces a visual calendar interface and checkbox-based food selection, which simplifies user interaction and reduces cognitive load compared to existing systems.

Framework

For the design of the FoodSaver application, it has been proposed to use the UserCentered Design framework. User-Centered Design is an iterative design process that focuses on understanding user

Needs, behaviors, and limitations throughout all design and development activities. Since food waste is largely affected by daily activities carried out in homes, such as food storage, shopping for food, and food preparation [1], [2], it is important to design a system based on actual user behavior rather than assumptions.

Studies suggest that food tracking, reminders about food expiration dates, and an intuitive interface can be useful tools for reducing food waste properly [3], [4]. In addition, persuasive system design principles proposed by B. J. Fogg emphasize that users' behavior can be changed through positive support from a system rather than forcing them to behave in a certain way, which aligns well with the objective of FoodSaver.

In order to effectively use User- Centered Design for system development, it is important to get in touch with actual users. To achieve this, face-to-face interviews will be carried out with people responsible for household food management (e.g., parents, students, domestic workers). This approach provides deeper insight into real-life behaviors, cultural practices, and everyday challenges related to food waste.

By adopting UCD, FoodSaver ensures usability, low cognitive load, cultural relevance within Saudi households, and a design that genuinely supports users in reducing food waste.

Collection Methods

The methods that this project will utilize include conducting semi- structured face-to-face interviews, followed by an online survey to obtain broader user feedback.

Method 1: Face-to-Face Interviews.

Purpose:

The purpose of this research is to understand household behaviors in food storage, cooking, grocery shopping, and waste disposal.

Participants:

The participants in this research will be conducted with individuals who are directly responsible for food management at home, These individuals include:

- Mothers
- Fathers who are responsible for grocery shopping.
- Individuals in the household who are responsible for cooking.
- Domestic workers who are responsible for food preparation.
- Students who live alone (including scholarship students abroad).
- Individuals who are interested in saving money and cutting down on waste.

This diversity ensures that we capture different perspectives and habits related to food management.

Implementation Steps:

1. Prepare a set of semi-structured questions, such as:
 - How do you usually manage food in your kitchen?
 - Who is responsible for buying groceries?
 - What are the main reasons food gets thrown away?
 - Do you check expiration dates regularly?
 - Would an app that reminds you about expiry dates be helpful?
2. Conduct 10-20 minute interviews face-to-face in a comfortable environment (such as the participant's home or a quiet meeting area).
3. Take written notes during the interview (and record audio if permission is granted).

4. Analyse responses by identifying repeated patterns and common problems, such as:

- Forgetting food in the refrigerator.
- Buying extra products during discounts.
- Cooking large quantities for guests.
- Lack of coordination between family members.

Tools Used:

- Printed Question Guide.
- Notebook for Observations.
- Voice Recorder.
- Google Forms for Survey.

Why Face-to-Face Interviews?

The following are some of the reasons why conducting face-to-face interviews would be useful:

- Observing Kitchen Environments.
- Understanding Family Dynamics.
- Unspoken Behaviours – Such as Storage Organization.

Method 2: Online Survey

After the interview analysis, a structured survey will be conducted to obtain quantitative data from a larger population. The survey will cover some of the common issues discussed in the interview. For example:

- Expiration tracking.
- Over-purchasing.
- Interest in reminder features.

This step confirms patterns and ensures the system design reflects broader user needs.

Survey Link:

https://docs.google.com/forms/d/e/1FAIpQLSfNf9RsZ7k9xLGbzZoI25VTp2mjIfLgSS_foc9J6_97zXhknQ/viewform?usp=header

Why Combining Interviews and Surveys?

By conducting interviews prior to surveys, one can delve into behaviors and challenges in-depth. By structuring the survey on the results of the interviews, one can ensure that the survey is relevant and targeted. By doing both, the validity of the research is improved and is in line with the User-Centered Design process, as FoodSaver is made according to user needs and Validated feedback.

User Analysis

The user analysis for FoodSaver is based on quantitative data collected through an online questionnaire involving 35 participants. The collected data ensured that the system design reflects real user behaviors, needs, and challenges related to household food management and food waste reduction.

User Profile: Household Food Managers

- **Demographics:** The primary users are individuals aged 18–24 (42.9%) and 35–44 (25.7%), representing young adults and family managers who are actively involved in household responsibilities.
- **User Types:** The main user groups consist of parents (42.9%) and individuals motivated by saving money and reducing food waste (37.1%).
- **Responsibilities:** A large majority of participants (82.8%) are responsible for grocery shopping and home cooking, making them the primary decision-makers in food management.

Users' Needs and Goals

- **Automated Tracking:** Since 48.6% of users reported throwing away food because it expired without their notice, an automated food-tracking system is essential.
- **Notification System:** 54.3% of participants identified expiration reminders as the most useful feature, highlighting the need for timely notifications to prevent waste.
- **Simplified Data Entry:** To reduce effort and cognitive load, 34.3% of users preferred a checkbox-based inventory system rather than manual text entry.
- **Culinary Inspiration:** 51.4% of users expressed the need for recipe suggestions based on available ingredients to solve the daily decision of what to cook.
- **User Goal:** The system aims to support the 40% of users who sometimes waste food and the 57.1% who do not regularly check expiration dates, helping them adopt more sustainable food management habits.

This user analysis will guide the design of FoodSaver, ensuring that all features and interactions are aligned with actual user needs and behaviors, following a user-centered design approach.

Task analysis

In consideration of the results obtained from the survey and in consideration of the basic concept behind the FoodSaver application, the general user tasks are as follows:

1. **Inventory Logging via Categories:**

Users are able to browse through different ingredient categories (vegetables, fruits, dairy products, meat products) and select available ingredients via checkboxes.

2. **Expiration Scheduling:**

Users are able to log in the application's calendar the expiration dates of products, allowing for better visualization of food freshness.

3. **Managing Smart Notifications:**

Users are able to receive notifications when products are nearing expiration dates, catering to the needs of 54.3% of surveyed users.

4. **Ingredient-Based Recipe Discovery:**

Users are able to select available ingredients, and the application automatically provides users with appropriate recipes.

5. **Social Interaction and Sharing:**

Users are able to share recipes, comments, and other culinary experiences via the application's community forum.

6. **Recipe Validation (Expert Task):**

Experienced mothers are able to review and approve submitted recipes before allowing users to access them.

20 scientific papers

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Food waste in households is one of the major kinds of waste and has a strong link to consumer behavior, including stock management and consumers' perceptions of expiration dates [1], [14], [15]. Research has shown that planning practices, storage behavior, and daily routines are important factors affecting food waste generation in households [2].

Several studies have identified potential intervention points for intelligent technologies such as food tracking technologies, expiration date notifications, and user interface design for reduced cognitive load [3]. Research on eco-feedback technologies has also shown that providing users with information about their consumption behavior can motivate them to engage in more sustainable practices [4], [11]–[13].

Moreover, persuasive design has also been identified as an important factor for motivating users to engage in behavioral change [5].

Food waste in Saudi Arabia has become a major economic concern. The economic cost of food waste has been estimated to be around 13 billion US dollars annually, which has a great impact on the national economy [6]. In addition, food waste levels in Saudi Arabia are reported to be around 427 kg per capita per year, which is beyond global averages and reflects consumers' behavior and lifestyle changes in urban areas [7].

Consumer misunderstanding of food labels, especially “best before” labels, has been recognized as a factor in the generation of avoidable food waste [6]. Culture and social behaviors, including those related to entertaining visitors, may also play a role in the generation of avoidable food waste [6], [7]. Technology, in the form of mobile phone applications, could potentially assist consumers in managing their food supplies and hence reduce avoidable purchases [6], [7], [11]–[13].

From a broader environmental perspective, avoidable food waste in Saudi Arabia is a key contributor to greenhouse gas emissions and resource use, including water usage [8], [9]. Methane, a potent greenhouse gas, is produced during the decomposition of avoidable food waste in landfills, with global warming potentials several times greater than carbon dioxide [9]. This underlines the need

for aligning efforts with broader national sustainability strategies, such as the Saudi Green Initiative [8], [9].

Recent studies have indicated that despite the improvement in awareness levels, there remains a difference between knowledge and consumer behaviour [10]. For example, the trend in impulse buying during promotional sales remains a driving force in consumer behavior. In this respect, the development of real-time technology in support of shopping management and household organization could contribute to the achievement of sustainability goals in the country, including the reduction of wastes as outlined in Vision 2030 [8], [10].

Recent advances in technology have been successful in illustrating the ability of intelligent applications in the reduction of food wastes in a household setting [16]. For example, the Exspiro mobile application has illustrated the significance of structured food tracking in the promotion of sustainable consumption behaviors in a household setting. The study has indicated the significance of timely food updates and statistical tracking in the reduction of food wastes. This aligns with FoodSaver's calendar-based tracking system.

Likewise, an artificial intelligence-based food consumption management application called Green Pantry utilizes behavioral models like the Motivation, Ability, and Opportunity model for analyzing the behavioral habits of users and reducing food waste [17]. In the research, the authors used a research-through-design approach by incorporating interviews, questionnaires, and usability testing for designing the application. The results showed high levels of user satisfaction with the application. The results also validated the design approach of the FoodSaver application by showing the effectiveness of using personalization and inventory management for improving the usability of the application. Furthermore, it also validated the approach of reducing cognitive load by using predefined categories of ingredients and check boxes.

Moreover, it has been demonstrated by researchers that artificial intelligence-based recipe recommendation systems using techniques like deep learning, knowledge graphs, and natural language processing can achieve high accuracy in identifying ingredients. It can also improve decision-making in cooking using computer vision. The results showed improved precision in recommending recipes [18]. The feature of the FoodSaver application for suggesting recipes using household ingredients can be improved using artificial intelligence.

Moreover, the limitation of matching the ingredients precisely has been overcome in the development of semantic-based recipe recommendation systems. This has been done through the introduction of substitute ingredient reasoning based on taste, texture, smell,[19] etc. The proposed model has been successful in achieving a higher degree of precision in recommending relevant recipes. This has reduced frustration in users when they are unable to find the required ingredients. The inclusion of similar reasoning in FoodSaver will help in increasing flexibility in the usage of food items rather than discarding them.

Lastly, intelligent grocery management systems based on IoT have been successful in managing food items effectively. This has been done through the tracking of inventory levels, expiration dates, and the creation of a shopping list. This has resulted in the reduction of household wastes [20]. In spite of the inclusion of hardware-based solutions in a few systems, the concept of automated inventory tracking aligns closely with the FoodSaver system.